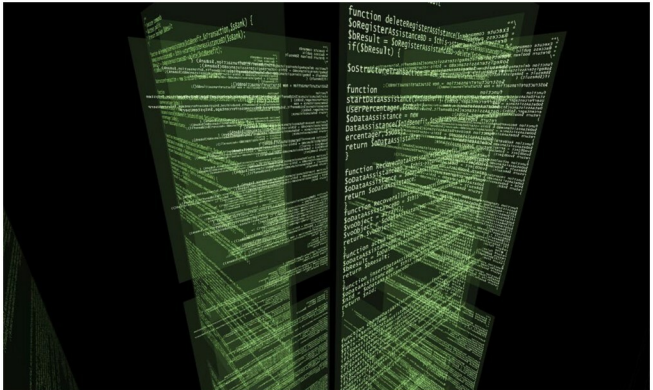


Apache OpenNLP: A Command Line Approach

Natural language processing (NLP) is a field of computer science, artificial intelligence and computational linguistics that explores the relationship between computers and natural languages. The Apache OpenNLP library is a machine learning toolkit, which processes natural language text written in Java. It supports most NLP tasks.



OpenNLP is an open source library for processing natural language texts using inbuilt machine learning tools. It is a toolkit from the Apache Software Foundation to support the most common NLP tasks, such as tokenisation, sentence detection and name finding. This library can be downloaded from the Apache mirror site <http://mirror.fibergrid.in/apache/opennlp/opennlp-1.5.3>. This demonstration uses OpenNLP version 1.5.3. The binary tar file *apache-opennlp-1.5.3-bin.tar.gz* is extracted and located under the */usr/local/apache-opennlp-1.5.3* directory of Ubuntu Linux.

It is necessary to download the pre-trained models for the OpenNLP 1.5 series from the site <http://opennlp.sourceforge.net/models-1.5/>. The following pre-trained models are used for tokenising, sentence detection and name finding in English language texts.

Tokenizer: Trained on OpenNLP training data
– *en-token.bin*

Sentence Detector: Trained on OpenNLP training data
– *en-sent.bin*

Name Finder: Location name finder model – *en-ner-location.bin*

Name Finder: Person name finder model – *en-ner-person.bin*

Name Finder: Organisation name finder model – *en-ner-organization.bin*

This is going to be demonstrated on an Ubuntu 12.04 LTS Linux system in this article.

Figure 1 shows the English language text file *inp_sentence.txt* used for testing the OpenNLP tools for the entire sequence of activity.

The first step

Test the command *opennlp* in Linux under the location */usr/local/apache-opennlp-1.5.3/bin*.

Tokenisation

Tokenizer segments input character sequences into tokens. Tokens are usually words, punctuation, numbers, etc.

The result of the SimpleTokenizer tool with input text file